Planning and Design Report

Android App: StudyMate - Student Planner and Progress Tracker

Introduction

This planning and design report defines the proposed Android application, StudyMate. The app is intended for students who need a simple way to manage academic tasks, study time, timetables, reminders, and marks. The design is based on the research findings from student planner and productivity apps, but it focuses on a realistic first version that can be built and tested in Android Studio.

The app will use Firebase Authentication for login, Firebase Cloud Firestore for cloud data storage, and a public quote API for a daily motivational message on the dashboard. The main purpose of the design is to make sure the app is clearly planned before development starts.

Brief Overview of the App



StudyMate will help students capture assignments, organise them by subject, view timetable entries, record marks, and monitor progress. The initial icon concept is a rounded calendar page with a check mark, using green and blue to represent calm organisation and academic progress. The app's innovative feature is a smart dashboard that combines deadlines, study progress, timetable items, and a daily motivational quote from an API. The quote feature is useful but not critical, so the app will still work if the API is unavailable.

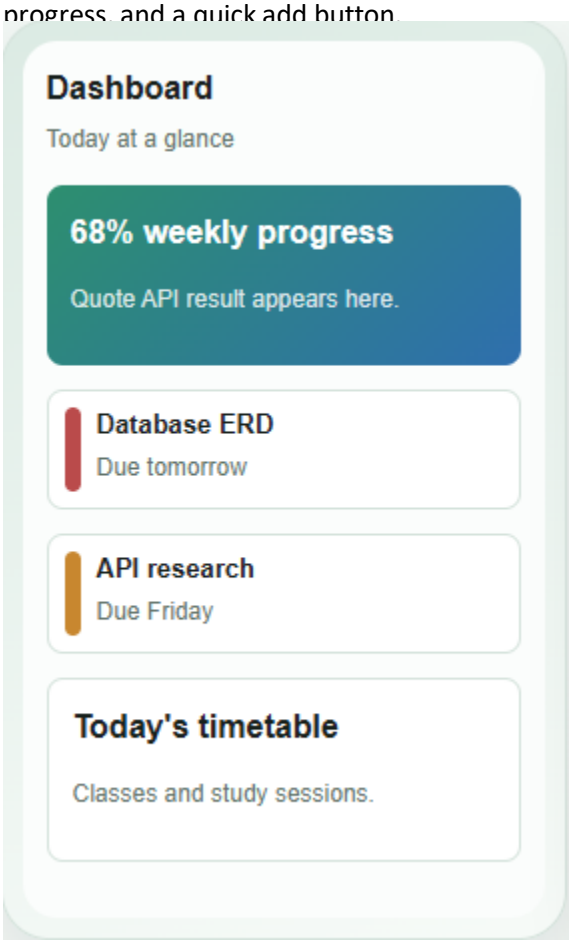
Detailed Requirements

- The user must be able to register, log in, and log out using Firebase Authentication.
- The user must be able to create, read, update, and delete subjects.
- The user must be able to create, read, update, and delete assignments.
- Each assignment must store a title, subject, description, due date, priority, completion status, and reminder setting.
- The user must be able to add timetable entries with subject, day, start time, end time, and location.
- The user must be able to record marks for assessments and view a simple average per subject.
- The dashboard must show upcoming deadlines, today's timetable items, weekly progress, and the daily motivational quote.
- The app must store user data in Firebase Cloud Firestore and separate each user's data by user ID.
- The app must use an external quote API and handle loading or network errors gracefully.
- The app must include unit tests for data validation, date sorting, and average calculation.

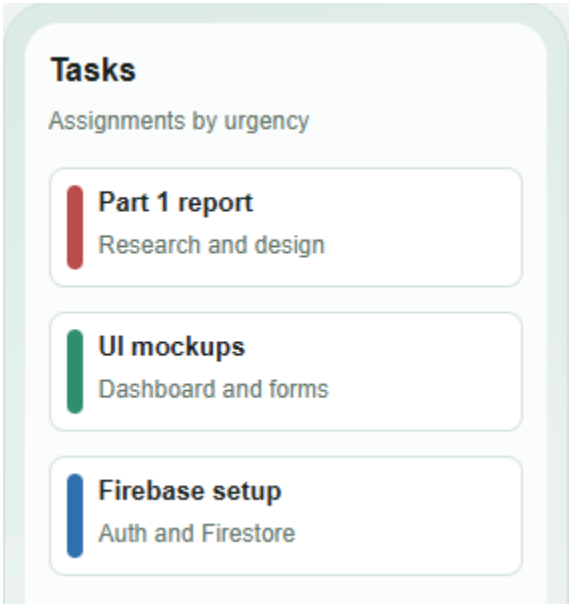
User Interface Design

The app will use a bottom navigation bar with four main screens: Dashboard, Tasks, Timetable, and Progress. A Settings screen will be available from the dashboard toolbar. The dashboard will be the home screen because it gives the user the quickest view of what matters today. The Tasks screen will show assignments grouped by due date and priority. The Timetable screen will show scheduled classes and study sessions by day. The Progress screen will show subjects, marks, and calculated averages.

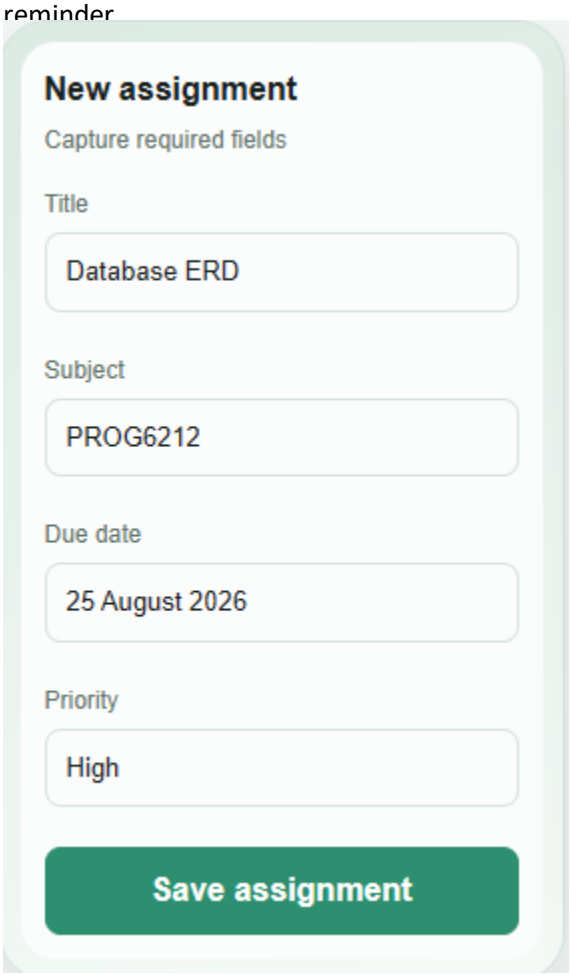
Dashboard screen: shows greeting, daily quote, today's classes, urgent assignments, weekly progress, and a quick add button.



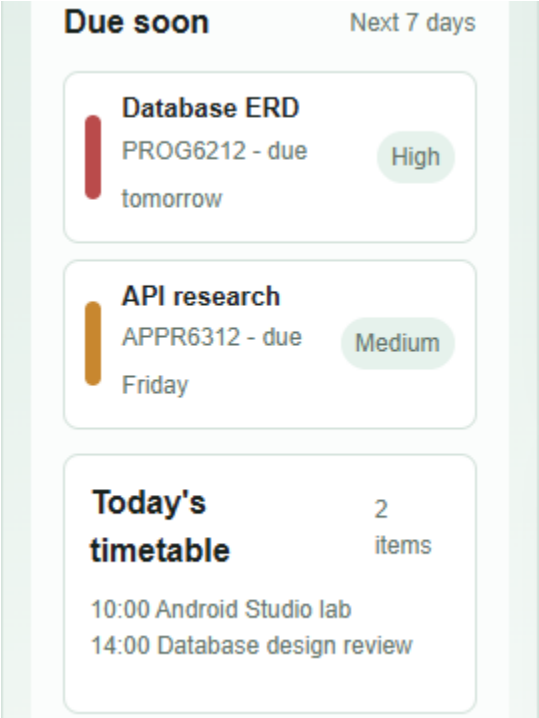
Tasks screen: shows assignment list with filters for all, due soon, and completed.



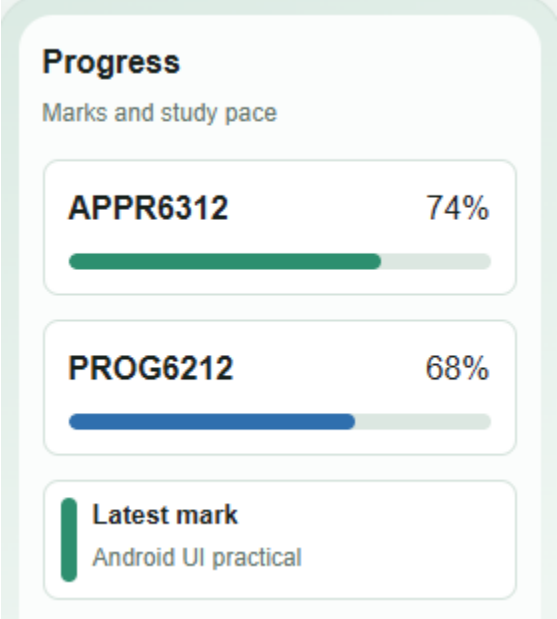
Add/Edit Assignment screen: captures subject, title, due date, description, priority, and reminder



Timetable screen: displays weekly class and study entries with add/edit actions.



Progress screen: displays subject averages and individual assessment marks.



- **Settings screen**: manages account logout and notification preferences.

API and Database Design

StudyMate will use Firebase Authentication for login and Firebase Cloud Firestore as the main database. Firestore is suitable because the app data is document-based and each user can have collections for

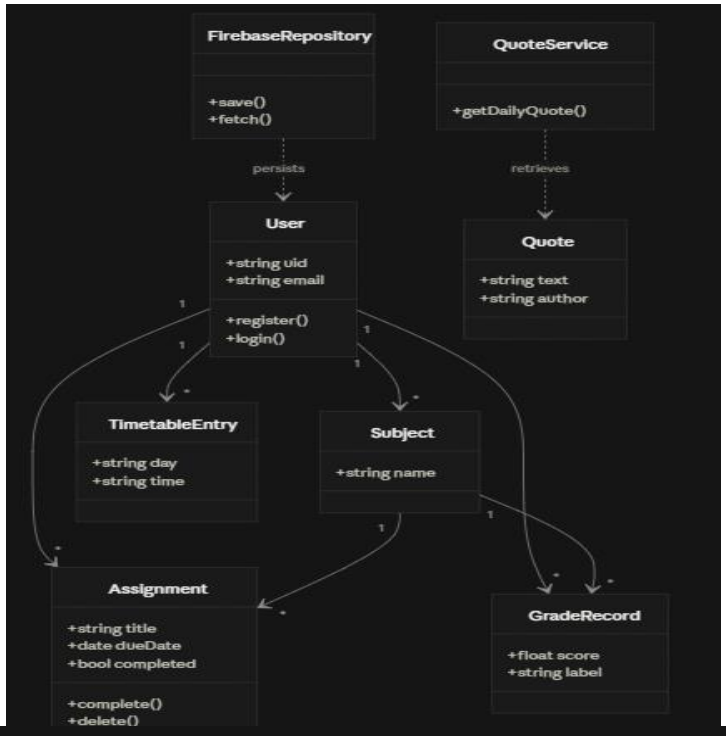
subjects, assignments, timetable entries, and grade records. The external API will be a public quote API such as Quotable or ZenQuotes. The app will send a GET request when the dashboard opens and display the quote text and author. If the API request fails, the dashboard will show a default local message so the app remains usable.

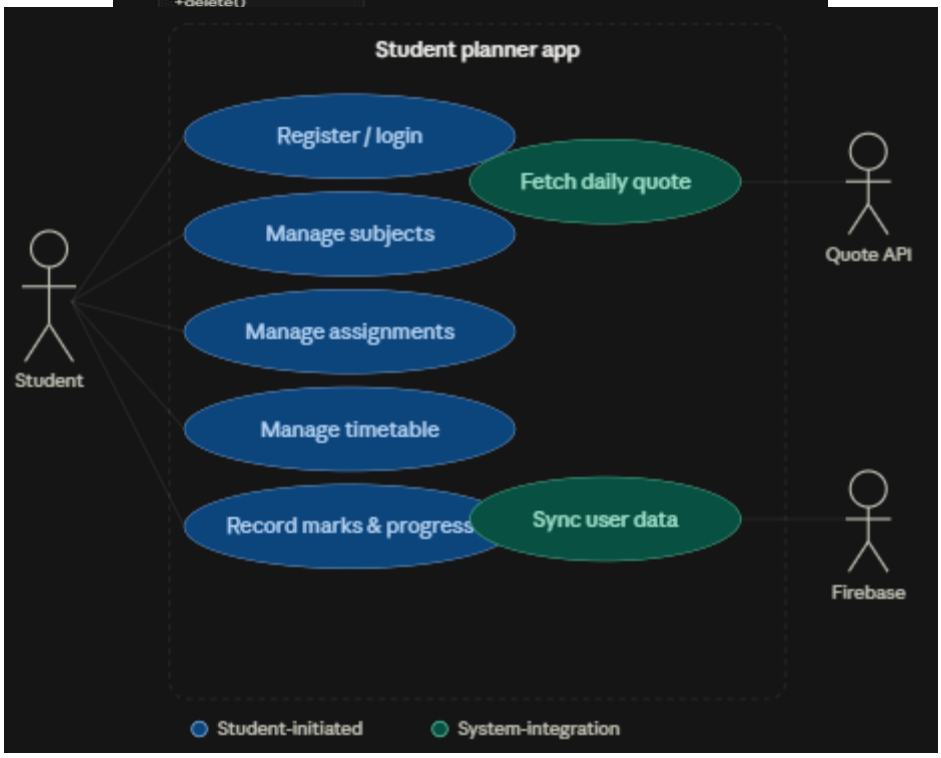
Firestore structure: users/{userId}/subjects/{subjectId}; users/{userId}/assignments/{assignmentId}; users/{userId}/timetable/{entryId}; users/{userId}/grades/{gradeId}. This structure keeps each student's data separate and makes it easier to test permission rules.

UML-Style Design

Use case summary:

- Student registers or logs in.
- Student manages subjects.
- Student adds, edits, completes, and deletes assignments.
- Student adds timetable entries.
- Student records marks and views progress.
- App fetches daily quote from external API.
- App stores and retrieves user data from Firebase.


FirebaseRepository
+save()
+fetch()
QuoteService
+getDailyQuote()
persists
retrieves
User
+string uid
+string email
+register()
+login()
Quote
+string text
+string author
TimetableEntry
+string day
+string time
Subject
+string name
Assignment
+string title
+date dueDate
+bool completed
+complete()
+delete()
GradeRecord
+float score
+string label


Student planner app
Register / login
Fetch daily quote
Manage subjects
Manage assignments
Manage timetable
Record marks & progress
Sync user data
Student
Quote API
Firebase
Student-initiated
System-integration

Data Captured and Stored

Data item	Fields	Data type examples
User	userId, email, displayName	String
Subject	subjectId, name, color	String
Assignment	assignmentId, subjectId, title, description, dueDate, priority, completed	String, Date, Boolean
Timetable entry	entryId, subjectId, dayOfWeek, startTime, endTime, location	String, Time
Grade record	gradeId, subjectId, assessmentName, mark, total, date	String, Number, Date

Project Plan

Week	Milestone	Main tasks	Testing focus
1	Research and planning	Complete app comparison, requirements, and UI design	Check requirements against brief
2	Project setup	Create Android Studio project, Firebase project, and repository	Build runs; GitHub Actions starts
3	Authentication and data models	Implement login, models, and repository classes	Unit tests for validation
4	Assignments and subjects	Build CRUD screens for subjects and assignments	Test create/edit/delete flows
5	Timetable and progress	Build timetable and grade tracker	Test date sorting and average calculation
6	API and dashboard	Connect quote API and finalise dashboard	Test API fallback and loading states
7	Polish and evaluation	Fix bugs, improve UI, prepare Play Store-ready assets	Run unit tests and final QA

Conclusion

StudyMate is designed as a focused student planner that can be developed within one project cycle. The requirements, screen plan, API design, database structure, UML-style summaries, and milestone plan provide a clear guide for development. By using Firebase for authentication and storage, a simple quote API for dashboard content, and focused unit tests for key calculations and validation, the app can meet the project brief while remaining realistic to build.

References

Firebase (2026) *Firebase Authentication and Cloud Firestore documentation*. Available at: https://firebase.google.com/docs (Accessed: 24 August 2026).

Quotable (2026) *Quotable API documentation*. Available at: https://github.com/lukePeavey/quotable (Accessed: 24 August 2026).

ZenQuotes (2026) *ZenQuotes API documentation*. Available at: https://zenquotes.io/ (Accessed: 24 August 2026).